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ETW3482

Week 10 Tutorial



This week's tutorial will have **two sessions**

1. **Quizzes for Week 10 revision.**
2. **A Hands-on practical session with questions.**



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What is the main purpose of honest assessment in predictive modelling?

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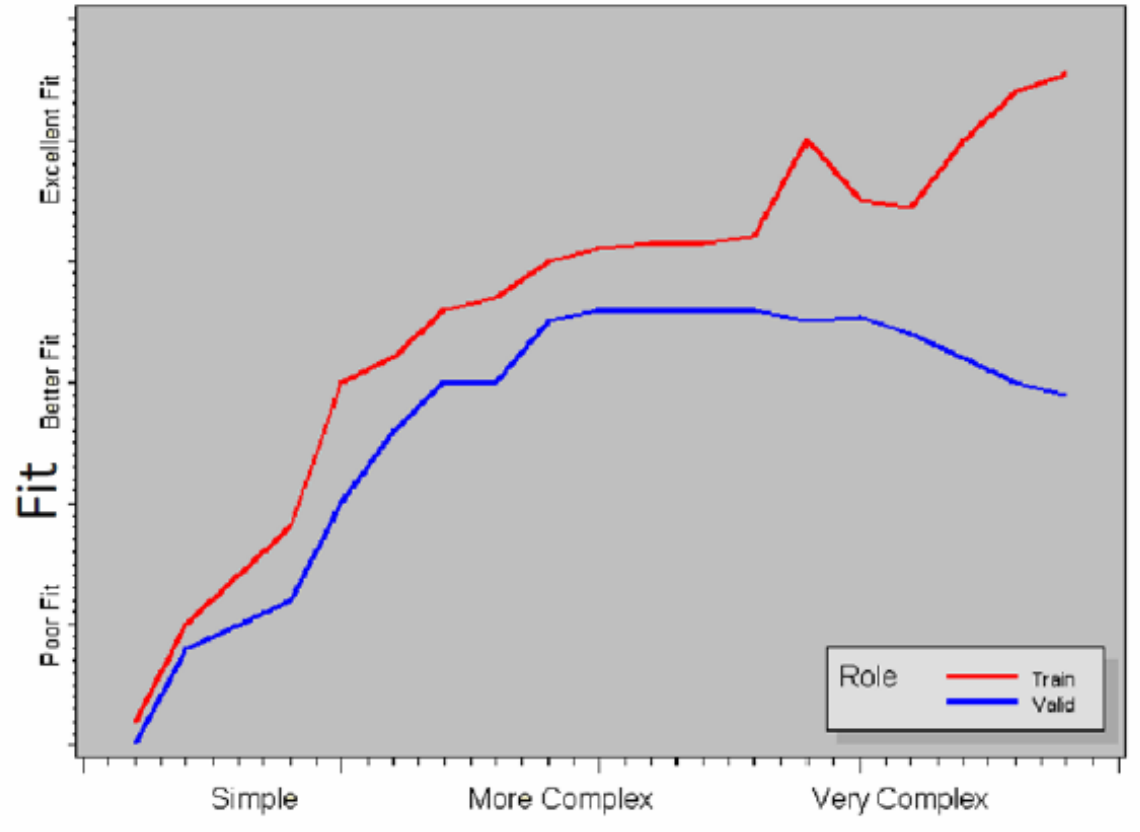
What does overfitting typically lead to in a predictive model?

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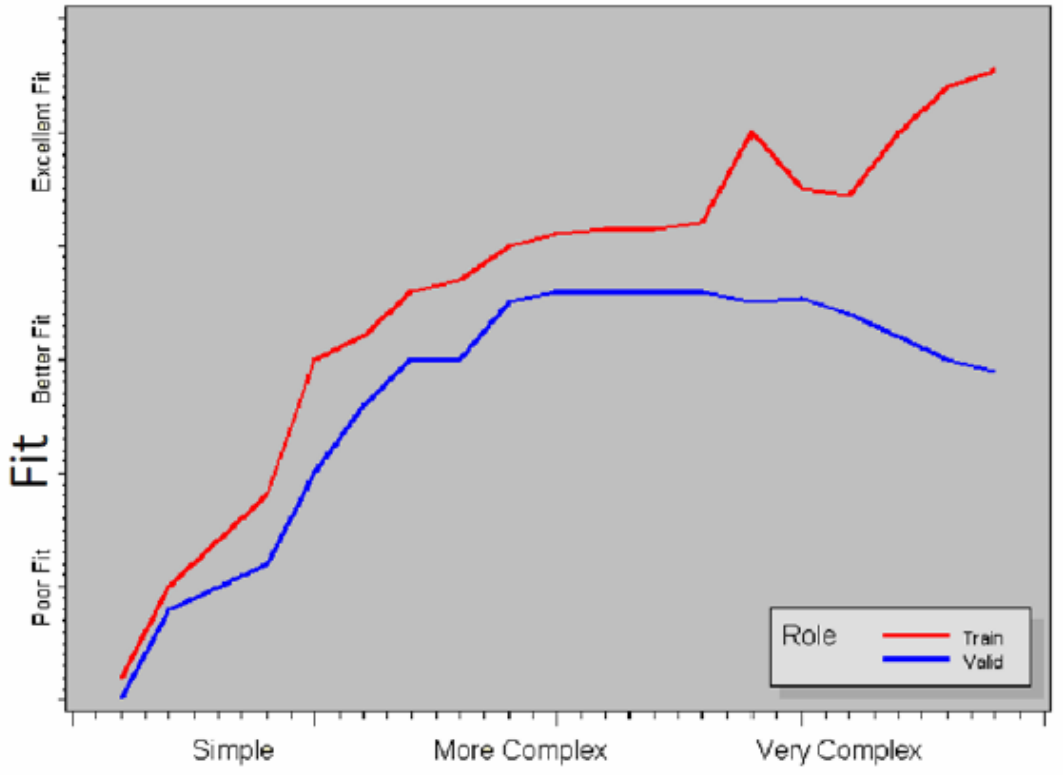
Fit versus Complexity

- Normally, model performance follows a straightforward trend.
- The fit on the training data improves as the complexity increases.
- After a certain degree of complexity, the fit improves due to the model identifying noises in the training data set.



Fit versus Complexity

- The overfitting patterns in training data are not likely to be repeated in the validation data or in future observations.
- Therefore, the models of varying complexity fits on the validation data,
- and the model with the complexity that has the highest validation fit statistic is selected.



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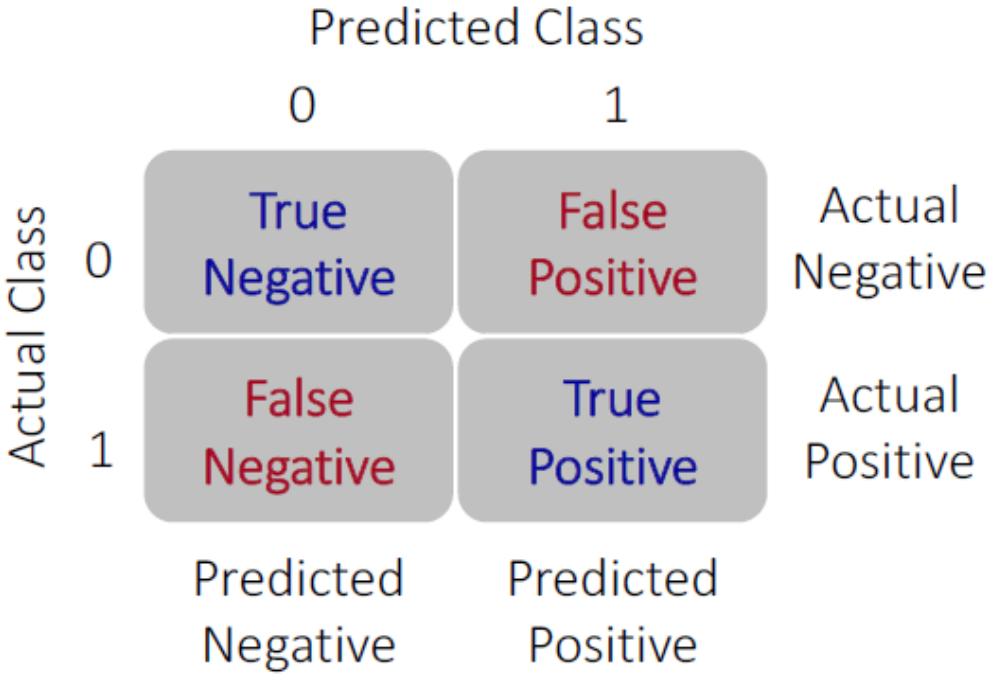
What does the confusion matrix primarily help evaluate?

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Confusion Matrix

- The confusion matrix is a crosstabulation of the actual and predicted classes.
- It quantifies the confusion of the classifier.
- The event of interest is often called a positive.



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Sensitivity = (i) = true positive / (ii)

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Specificity = (i) = (ii) / (iii)

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Which performance metric measures the proportion of actual positives correctly predicted?

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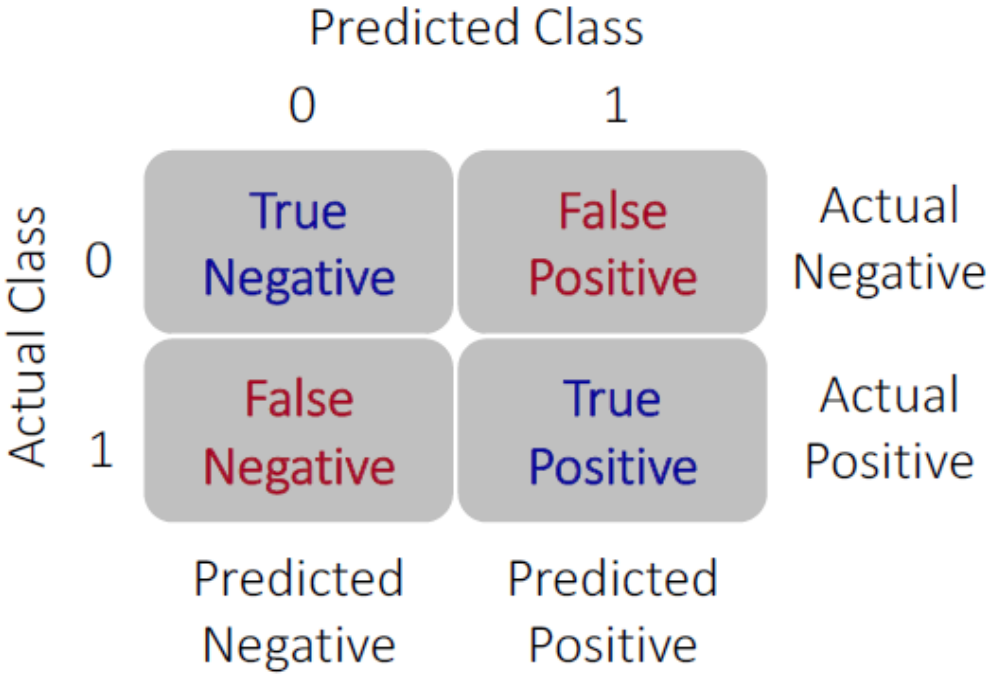
Confusion Matrix

- The simplest performance statistics are **accuracy**,

$$\text{accuracy} = \frac{\text{true positive and negatives}}{\text{total cases}}$$

- and **error rate or misclassification rate**,

$$\text{error rate} = \frac{\text{false positive and negatives}}{\text{total cases}}$$



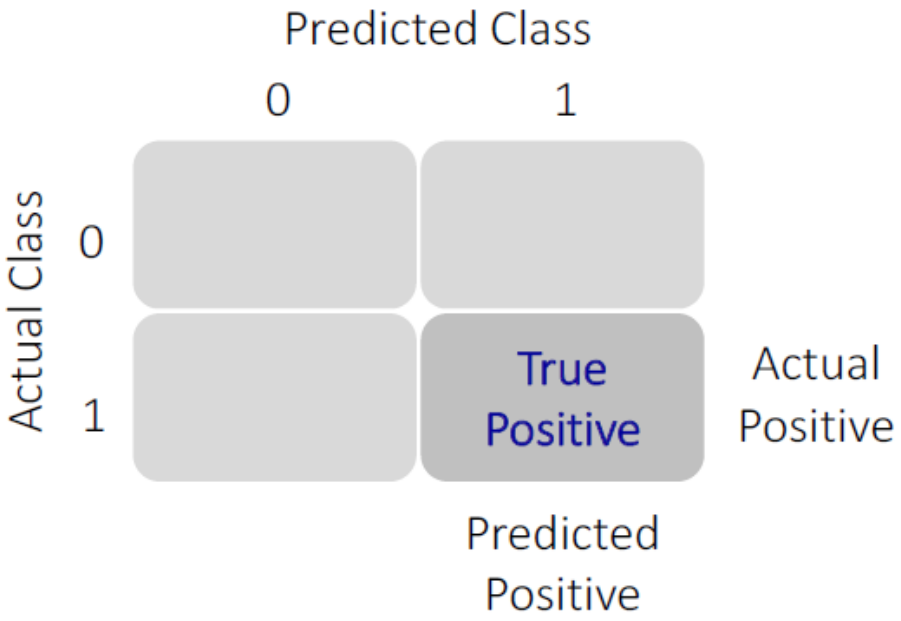
Confusion Matrix

- Two specialized measures of classifier performance are **sensitivity**,

$$\text{sensitivity} = \frac{\text{true positives}}{\text{total actual positives}}$$

- and **positive predicted value** (PV+),

$$\text{positive predicted value} = \frac{\text{true positives}}{\text{total predicted positives}}$$



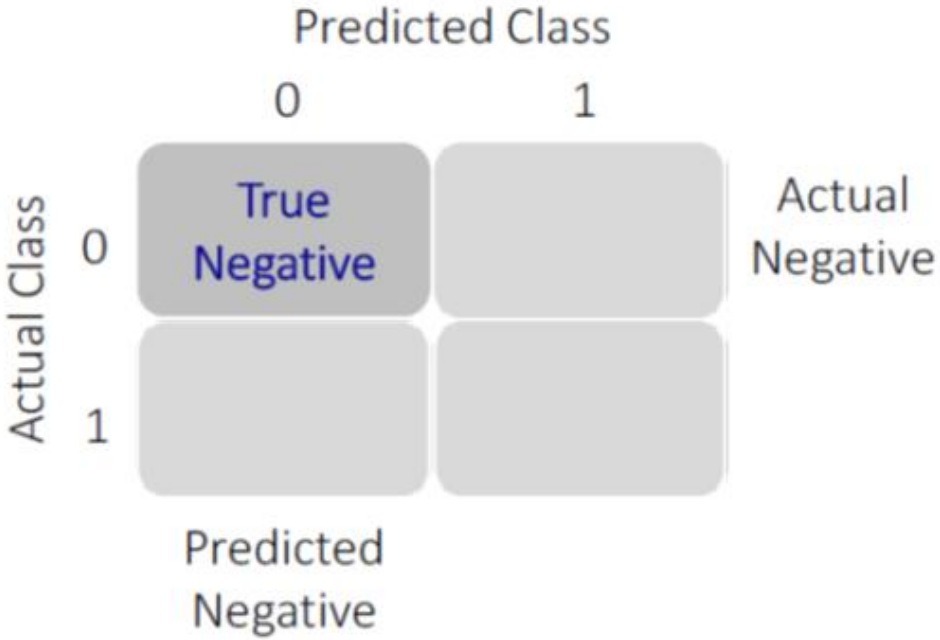
Confusion Matrix

- The analogs to these measures for true negatives are **specificity**,

$$\text{specificity} = \frac{\text{true negatives}}{\text{total actual negatives}}$$

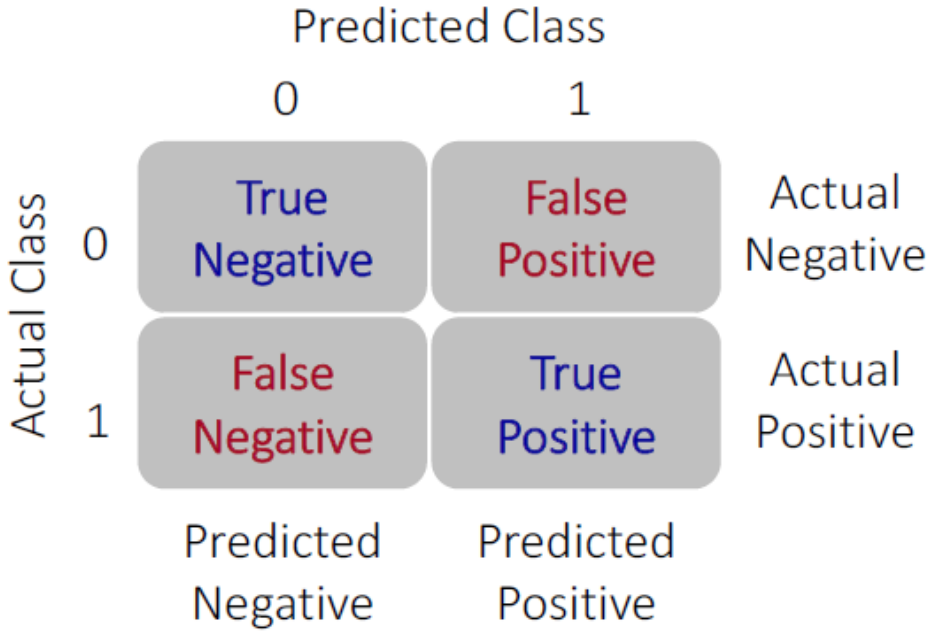
- and **negative predicted value** (PV-),

$$\text{negative predicted value} = \frac{\text{true negatives}}{\text{total predicted negatives}}$$



Confusion Matrix

- Ideally, you want large values for all these statistics.
- The context of the problem determines which of these measures is the primary concern.



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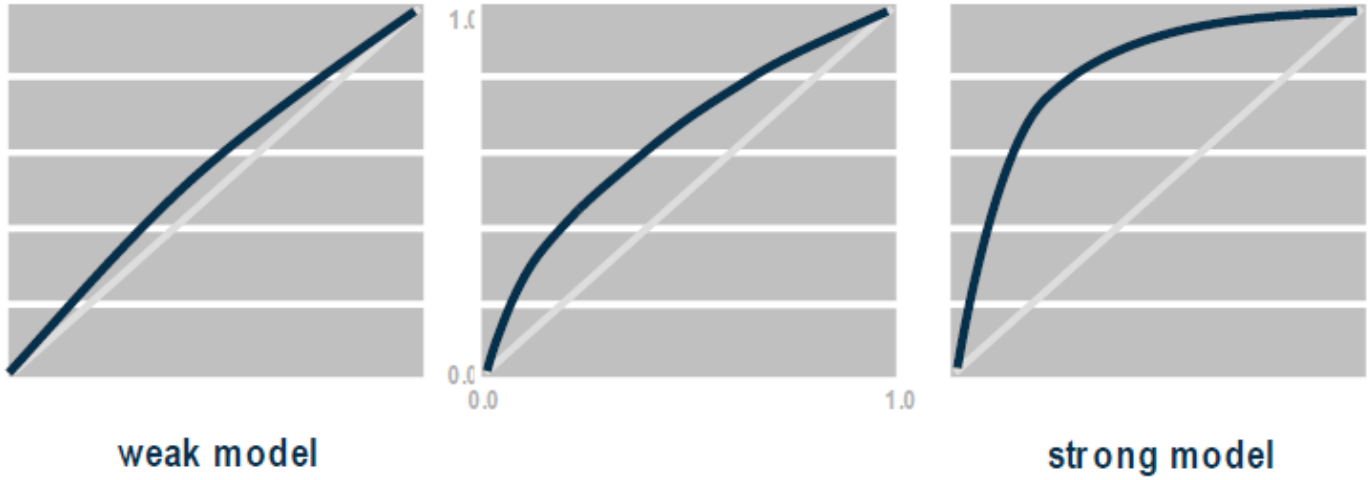


What does the area under the ROC curve represent?

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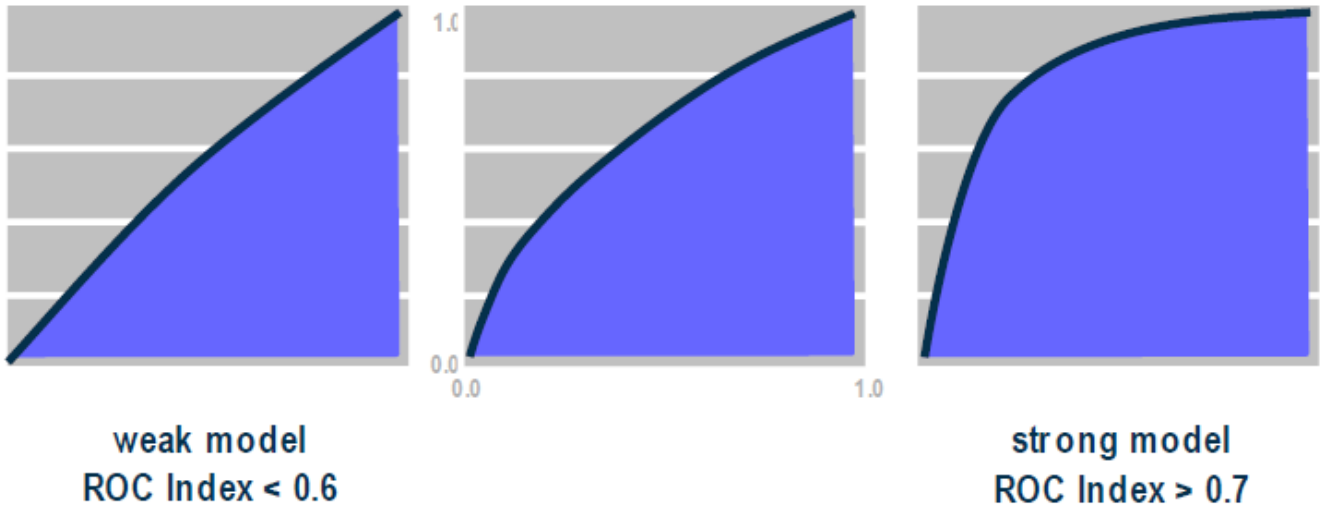
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ROC Curve



- As the ROC curve bows above the diagonal, the predictive power increases.
- Models that capture mostly primary outcome cases without capturing secondary outcome cases are strong models.
- A perfect model would reach the (0, 1) point where both sensitivity and specificity equal 1.

ROC Curve



- The tradeoff between primary and secondary case capture can be summarized by the area under the ROC curve.
- In SAS, this area is called the ROC index. In statistics, it is more commonly called the *c*-statistic.

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Which statistic is commonly used for ranking prediction in model comparison?

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What is the average squared error (ASE) used to measure?

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Which of the following applications is most likely to prioritise sensitivity as the primary performance measure?

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What is the most appropriate performance metric for evaluating a spam email filter?

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Why is positive predictive value (PV+) important in a targeted marketing model?

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Which is true for this cumulative lift chart?

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Which is true for the cumulative response chart?

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PART 2:
PRACTICAL HANDS-ON SESSION



Practical session

1. Refer to the week 10 demo guide.
2. Follow the tutor’s instructions.

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WRAP UP



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Q&A



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Audience Q&A

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